

Copilot Instructions - Centro de Consulta de Acuerdos Sanitarios

Project Overview

This is a Mexican government web application for managing cattle health agreements between two specialized working groups:

- **CESO** - Consejo Estatal de Seguimiento Operativo del SINIDA (State-level cattle traceability via ear tags)
- **APHIS-USDA Working Group** - Federal/international cooperation for bovine tuberculosis control and cattle export certification

The app serves as an **agreement monitoring system** and **document repository** for critical animal health documentation.

Architecture & Technology Stack

Frontend: React 18 + Vite, Bootstrap 5, government design framework (gob.mx)

Backend: Firebase (Firestore, Storage, Auth, Hosting)

Data: Excel databases imported to Firestore collections

Styling: Official Mexican government design system with custom CSS variables

Key Architectural Patterns

```
// Dual collection structure for agreements
firestore/
├─ acuerdos-ceso/           # CESO agreements
├─ acuerdos-aphis/         # APHIS-USDA agreements
└─ {collection}/{docId}/evidencias/ # Evidence subcollections
```

Agreement Data Model (`src-development/models/dataModels.js`):

- Unified structure for both CESO and APHIS agreements
- Excel import/export compatibility with date parsing

- Status normalization: Pendiente , Completado , Vencidos
- Auto-detection of source organization by agreement number patterns

Critical Development Workflows

Firebase Security Model

```
// Public read access for agreements listing
// Authenticated write access only
// Evidence files: public metadata, auth-required uploads
```

Key Commands

```
# Development with emulators
npm run emulators:start
npm run test:emulator

# Deployment (production is live at tb-yucatan.web.app)
npm run deploy           # Full deployment
npm run deploy:hosting    # Frontend only
npm run deploy:rules      # Security rules only
```

Service Account Security

- **NEVER commit** Firebase service account keys
- Store in `C:\Users\%USERNAME%\FirebaseKeys\` outside project
- Use environment variables for paths in `.env` files
- Rotate keys every 90 days per government security standards

Project-Specific Conventions

Government Design System

```
:root {  
  --color-gobierno-principal: #611232;  
  --color-gobierno-secundario: #9d2449;  
  --color-gobierno-dorado: #a57f2c;  
}  
/* Use .btn-gobierno-*, .card-gobierno, .table-gobierno classes */
```

File Structure Patterns

- src-development/ - React source code being reconstructed
- static/ - Compiled production assets with hash names
- BASES DATOS/ - Excel databases (usuarios_aphis-usda_new_web_app.xlsx, usuarios_ceso_new_web_app.xlsx)
- Bilingual support throughout (Spanish primary, English secondary)

Role-Based Access Control

```
// USER_ROLES: PUBLIC, RESPONSIBLE, ADMINISTRATOR  
// Permissions: view_agreements, upload_evidence, mark_completed, manage_users  
// Auth context in src-development/context/AuthContext.jsx
```

Integration Points & Dependencies

Excel Data Import Pattern

```
// Agreement number patterns determine source organization  
// CESO: specific format patterns  
// APHIS-USDA: different numbering convention  
// Parse Excel date serial numbers to JavaScript Date objects
```

Evidence Management

- Firebase Storage paths: acuerdos-{source}/{docId}/evidence-files/

- Firestore metadata: {collection}/{docId}/evidencias/{evidId}
- Supported formats: PDF, JPG, PNG, Word documents
- Public viewing, authenticated upload workflow

Firestore Emulator Testing

```
// tests/emulator-e2e.js - End-to-end testing with local emulators
// Validates Firestore rules, storage access, authentication flows
```

Specialized Domain Knowledge

Mexican Government Context

- Uses official gob.mx framework for styling and accessibility
- Follows SENASICA (animal health service) protocols
- Integrates with SINIDA cattle identification system
- Supports international certification workflows with USDA

Cattle Health Terminology

- **Acuerdos:** Official agreements/resolutions from working group meetings
- **Evidencias:** Supporting documentation/proof of agreement compliance
- **Seguimiento:** Follow-up/monitoring of agreement status
- **Semovientes:** Live cattle (legal term for livestock in transit)

Common Patterns to Follow

1. **Bilingual Support:** Always provide Spanish and English text options
2. **Excel Compatibility:** Maintain date formats and column structures for database imports
3. **Government Branding:** Use official colors, fonts (Patria/Noto Sans), and layout patterns
4. **Security First:** Never expose authentication tokens, always validate user permissions
5. **Mobile Responsive:** Government accessibility requirements mandate mobile-first design

When working with this codebase, prioritize understanding the dual-organization structure (CESO vs APHIS) and the evidence management workflow, as these are the core business logic patterns that drive most features.